

Juan David Muñoz-Bolaños

Optical System Engineer

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Summary

Hybrid-profile **Optical Systems Engineer** and **Machine Vision Developer** with 7 years of experience bridging deep physics R&D with industrial series production. Proven track record in delivering high-speed Halcon vision pipelines for quality control and building complex 3D metrology systems from scratch. Currently finalizing a PhD in Adaptive Optics focused on real-time wave-field control and hardware-software integration.

Professional Experience

Mar 2018 to Mar 2019 **INTECOL S.A.S.**

Medellín, Colombia

Junior Machine Vision System Engineer

High-Speed Industrial Inspection: Engineered and deployed Halcon-based vision systems for high-speed glass labeling lines. Designed custom dual-mode (bright-field/dark-field) LED illumination for 100% reliable detection of label edges, liquid levels, and cap sealing states; achieved full industrial client sign-off.

Robotic Integration (Renault): Implemented the Cognex VisionPro stack for the automated recognition and pose estimation of automotive plastic parts, enabling a robotic arm to perform precision surface flame treatment.

System Validation: Selected and calibrated industrial cameras, lenses, and specialized lighting; performed on-site system qualification (Gage R&R) on live production lines.

Jan 2022 to Present **Medical University of Innsbruck**

Innsbruck, Austria

PhD Researcher | Adaptive Optics & Metrology

Real-Time Wavefront Control: Built a fully automated adaptive optics stack using a 63-actuator deformable mirror and scientific sCMOS cameras. Developed the complete C++/Python/LabVIEW control architecture for real-time phase retrieval and iterative aberration correction.

3D Metrology & Simulation: Characterized nanometer-scale aberrations via Zernike decomposition and wave-field modeling in Python. Performed systematic ZEMAX OpticStudio tolerance analysis to ensure diffraction-limited resolution (0.3 μm lateral).

System Integration: Led the end-to-end hardware-to-software build of a multiphoton microscope system, managing laser source integration (fs/CW) and high-speed data acquisition (ACS Photonics '25).

Jun 2020 to Dec 2021 **Leibniz IPHT**

Jena, Germany

M.Sc. Researcher | Hyperspectral AI & Photonics

Spectrometer Prototyping: Led the optical layout, detector calibration, and construction of a dispersive 1064 nm fiber-probe Raman spectrometer for non-invasive tissue analysis (Applied Sciences '24).

Hyperspectral Data Analysis: Developed automated AI pipelines (clustering, supervised ML) using the custom SpectraMap Python package to classify cancerous vs. healthy tissue cells with high accuracy.

Technical Competencies

Image Processing & Computer Vision: Halcon (MVTec) · Cognex VisionPro · OpenCV · Image segmentation · Defect detection · 3D Metrology · Illumination design (bright-field, dark-field, structured

light).

AI & Machine Learning: Supervised ML (Python, NumPy, scikit-learn) · Clustering and classification models · SpectraMap package · Pattern recognition for automated quality control.

Software Development: **Python (advanced)** · Halcon (intermediate) · C++ (intermediate; OpenCV, real-time control) · LabVIEW/FPGA · Git · MATLAB · C#/.NET awareness.

Precision Metrology: Wavefront sensing · Zernike decomposition · ZEMAX OpticStudio (tolerancing) · Aberration correction · Laser alignment (fs, CW, 1064 nm).

Industrial Integration: Hardware-software integration of camera systems and laser modules · Supplier management (Thorlabs, Hamamatsu) · EH&S lab safety · ISO standard awareness.

Education

Jan 2022 to Present	Medical University of Innsbruck <i>PhD in Adaptive Optics & Microscopy</i>	Austria
2019 to 2021	Friedrich Schiller University Jena <i>M.Sc. in Photonics</i>	Germany
2012 to 2018	University of Cauca <i>B.Sc. in Physics Engineering</i>	Colombia

Publications & Awards

Awards: ASML & ZEISS Summer Schools (Selected); COLFUTURO Grant 2026 Awardee.

Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. ACS Photonics 12(1), 176–184 (2025).

Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. Applied Sciences, 14(11), 4726 (2024).

Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. Biomedical Optics Express 14(4), 1562–1578 (2023).

Languages

English (Professional) · German (B2, ongoing) · Spanish (Native)

Soft Skills

Communication & Collaboration: Presentation of complex technical results through peer-reviewed publications, conferences (SPIE Photonics West), and international seminars within cross-functional teams.

Problem Solving: Proven ability to resolve complex, cross-disciplinary challenges, evidenced by the successful delivery of three peer-reviewed publications (ACS Photonics, Biomedical Optics Express, Applied Sciences) and international conference proceedings.

Adaptability: Successfully navigated a global career path from Colombia to Germany and Austria. Demonstrated high adaptability and networking skills by building professional connections and integrating into diverse work cultures, research environments, and technology stacks.

Hobbies

Mountain biking, dancing Bachata, playing harmonica, learning German.

References

Prof. Monika Ritsch-Marte
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Medical University of Innsbruck

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Dr. Christoph Krafft

Spectroscopy Group Leader

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Leibniz-IPHT Jena